

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2383

Bonwood 230 kV
January 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2383. The Interconnection Customer proposes a 24.0 MW natural gas combined cycle facility interconnecting at the Bonwood 230 kV substation in the Ohio area. The request was received on February 17, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 534 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$172,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CLAUS – UNION 138 kV line
Contingency: NSP – WISSOTA 138 kV
Loading: 112.0% (Rating: 434 MVA)
DFAX: 5.0%
Pre-project loading: 111.7%
2. Facility: CLAUS – SAN ANDREAS 138 kV
Contingency: WISSOTA – NSP 138 kV
Loading: 111.8% (Rating: 454 MVA)
DFAX: 5.0%
Pre-project loading: 111.5%
3. Facility: SAN ANDREAS – WISSOTA SUBSTATION 138 kV
Contingency: WISSOTA – NSP 138 kV
Loading: 111.7% (Rating: 459 MVA)
DFAX: 5.0%
Pre-project loading: 111.5%
4. Facility: BERRY ST 345/138kV Auto
Contingency: RUSSELLVILLE EAST – RANCHO CARMEL 138 kV
Loading: 110.4% (Rating: 647 MVA)
DFAX: 7.8%
Pre-project loading: 110.1%
5. Facility: WHARTON – TOLEDO BEND SUBSTATION 230 kV
Contingency: BARRETT SUBSTATION – NORWICH 500 kV line
Loading: 102.1% (Rating: 897 MVA)
DFAX: 11.2%
Pre-project loading: 101.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CLAUS – UNION 138 kV line	Rebuild line with higher-capacity conductor	\$2,104,000	0% (\$10,000)
CLAUS – SAN ANDREAS 138 kV	Rebuild line with higher-capacity conductor	\$7,043,000	0% (\$33,000)
SAN ANDREAS – WISSOTA SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$9,549,000	0% (\$45,000)
BERRY ST 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$14,519,000	0% (\$71,000)
WHARTON – TOLEDO BEND 230 kV	Reconductor line	\$2,999,000	0% (\$13,000)
	Total allocated to project		\$172,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.